55 x 40 mm target | CM5 / CM5 Lite | USB-C 5 V input + USB0 recovery | single USB3 host | dual 4-lane MIPI | GPIO | SD | fan | RTC

New physical parts and reviewed contact mappings. Prototype verification and PCB layout remain required. Do not order from schematic alone.

CM5 module contacts / GPIO voltage selection / controls

C1 10uF 5V

H2: fit shunt 2–3 for default 3.3V; 1–2 selects 1.8V. Power OFF before changing.
GPIO bank follows VREF; header 3.3V pins remain 3.3V.
MIPI1 / HAT I2C follows GPIO_VREF; verify peripherals in 1.8V mode.

3.3 V peripheral regulator — sequenced by CM5_3V3

C3 22uF 5V
C4 100nF GND
R25 CM4_3V3 PWR_EN
C17 100nF GND

NetC2_2 R22 NetR2_2
C2 1uF NetC2_1 NetC2_2
L1 3.3uH NetC2_1 MIPI_3.3V

MIPI_3.3V R23 FB1:0.8V
C5 15pF FB1:0.8V MIPI_3.3V
R24 FB1:0.8V GND
R26 NetR6_1 FB1:0.8V

U1 MP1477GTF-Z
5V 1 VIN
PWR_EN 5 EN
GND 3 GND
2 SW NetC2_1
4 BST NetR2_2
6 FB NetR6_1

Fan and RTC

C19 1uF RTC_VDD GND
C23 10uF 5V GND
C24 10uF FAN_5V

U11 RT9742GGJ5
5V 5 IN
NetR7_2 4 EN
GND 2 GND
1 OUT FAN_5V
3 FLG NC

R27 VBUS_EN NetR7_2

USB-C power + recovery

BOOT: hold during power-up for USB recovery.
POWER: module power button. CM5 eMMC uses onboard storage; external microSD applies only to CM5 Lite.

Key1 PI_BOOT SJU-3000P GND
Key2 PWR_BUTTON SJU-3000P GND

USB3 SuperSpeed ESD

U2 TPD4EUSB30DQAR
USB3-0-TX_N 10 NC
USB3-0-TX_N 1 I/O1
USB3-0-TX_P 2 I/O2
USB3-0-TX_P 9 NC
3 GND
8 GND
7 NC USB3-0-D_N
4 I/O3 USB3-0-D_N
5 I/O4 USB3-0-D_P
6 NC USB3-0-D_P

C6 22uF MIPI_3.3V
C16 100nF MIPI_3.3V
C14 1uF MIPI_3.3V
C15 1uF MIPI_3.3V
C7 22uF MIPI_3.3V

H3 RTC_VDD 1 RTC_VDD 2 GND SHIELD MP GND
BM02B-SRSS-TB(LF)(SN)

H4 FAN_TACHO 4 FAN_TACHO
FAN_PWM GND 3 GND
FAN_PWM
FAN_5V 1 FAN_5V
SHIELD MP GND
BM04B-SRSS-TB(LF)(SN)

Host power and Indicators

U3 RT9742CGJ5
5V 5 IN
VBUS_EN 4 EN
GND 2 GND
1 OUT USB_5V
3 FLG NC

U15 74LVC1G07SE-7
CM4_3V3 5 VCC
PI_PWR_LED 2
3 GND
4 PWR_LED_SINK

C25 100nF CM4_3V3 GND
C9 10uF USB_5V GND
C10 10uF 5V GND

CM5 Lite microSD supply

C11 10uF CM4_3V3
C12 10uF SD_3.3V GND
C13 100nF SD_3.3V
R15 SD_PWREN CM4_3V3
R12 SD_CLK NetR12_2
PWR_LED_SINK D1 Red NetLed1_1 LED_G D2 Green NetLed1_2

U4 RT9742GGJ5
CM4_3V3 5 IN
SD_PWREN 4 EN
GND 2 GND
1 OUT SD_3.3V
3 FLG NC

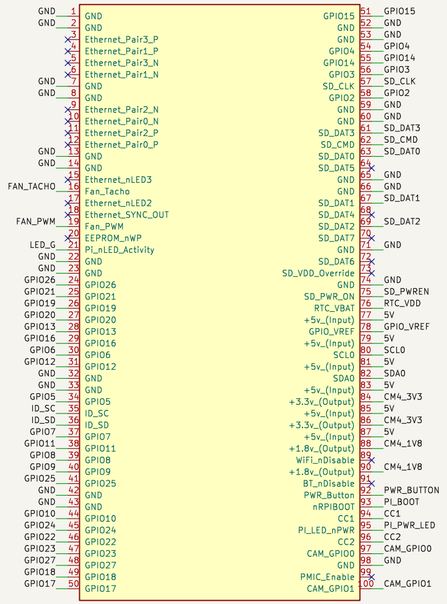
R10 CM4_3V3 NetLed1_1
R11 NetLed1_2 CM4_3V3

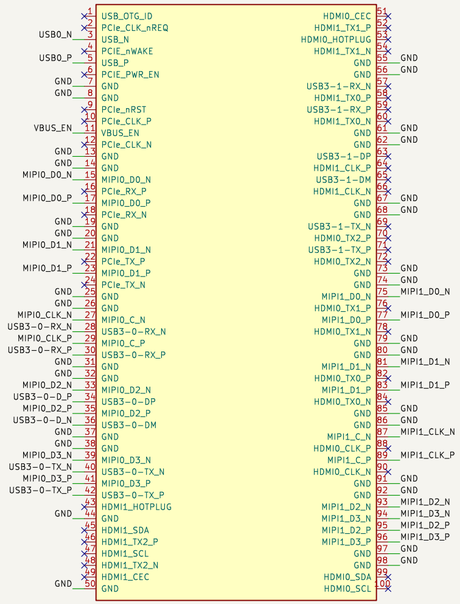
J2 692122030100
1 VBUS USB_5V
2 D- USB3-0-D_N
3 D+ USB3-0-D_P
5 SSRX- USB3-0-RX_N
6 SSRX+ USB3-0-RX_P
8 SSTX- USB3-0-TX_N
9 SSTX+ USB3-0-TX_P
SHIELD DRAIN GND
SH 7 4
GNDGND

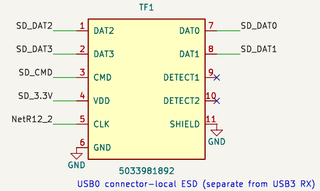
CM1
GND 1 GND
GPIO15 21 GPIO15

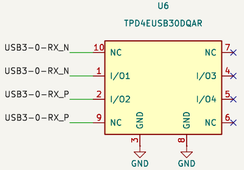
CM2
1 USB_OTG_ID
HDMI0_CEC 21

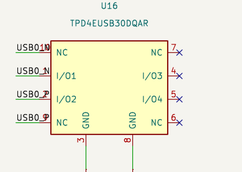
TF1


CM1
GND 1
GND 2
Ethernet_Pair3_P 3
Ethernet_Pair1_P 4
Ethernet_Pair3_N 5
Ethernet_Pair1_N 6
GND 7
GND 8
Ethernet_Pair2_N 9
Ethernet_Pair0_N 10
Ethernet_Pair2_P 11
Ethernet_Pair0_P 12
GND 13
GND 14
Ethernet_nLED3 15
FAN_TACHO 16
Fan_Tacho 17
Ethernet_nLED2 18
Ethernet_SYNC_OUT 19
FAN_PWM 20
Fan_PWM 21
EEPROM_nWP 22
LED_G PI_nLED_Activity 23
GND 24
GND 25
GPIO26 26
GPIO21 27
GPIO19 28
GPIO20 29
GPIO13 30
GPIO16 31
GPIO6 32
GPIO12 33
GND 34
GND 35
GPIO5 36
ID_SC 37
ID_SD 38
GPIO7 39
GPIO11 40
GPIO8 41
GPIO9 42
GPIO25 43
GND 44
GND 45
GPIO10 46
GPIO24 47
GPIO22 48
GPIO23 49
GPIO27 50
GPIO18 51
GPIO17 52
GPIO15 61
GND 62
GND 63
GPIO4 64
GPIO14 65
GPIO3 66
SD_CLK 67
GPIO2 68
GND 69
GND 70
SD_DAT3 71
SD_CMD 72
SD_DAT0 73
SD_DAT5 74
GND 75
GND 76
GND 77
SD_DAT1 78
SD_DAT4 79
SD_DAT2 80
SD_DAT7 81
GND 82
SD_DAT6 83
SD_VDD_Override 84
GND 85
SD_PWREN 86
RTC_VBAT 87
5V 88
GPIO_VREF 89
SCL0 90
5V 91
SDA0 92
5V 93
CM4_3V3 94
5V 95
CM4_3V3 96
5V 97
CM4_1V8 98
99
CM4_1V8 100
PWR_BUTTON 101
PI_BOOT 102
CC1 103
PI_PWR_LED 104
CC2 105
CAM_GPIO0 106
GND 107
108
CAM_GPIO1 109
SD_PWR_ON 110
RTC_VDD 111
+5v_(input) 112
+5v_(input) 113
+3.3v_(Output) 114
+5v_(input) 115
+3.3v_(Output) 116
+5v_(input) 117
+1.8v_(Output) 118
WiFi_nDisable 119
+1.8v_(Output) 120
BT_nDisable 121
PWR_Button 122
nRPIBOOT 123
124
10164227-1004A1RLF

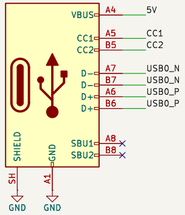

CM2
USB_OTG_ID 1
PCIe_CLK_nREQ 2
USB0_N USB_N 3
PCIE_nWAKE 4
USB0_P USB_P 5
PCIE_PWR_EN 6
GND 7
GND 8
PCIe_nRST 9
PCIe_CLK_P 10
VBUS_EN 11
VBUS_EN PCIe_CLK_N 12
GND 13
GND 14
MIPI0_D0_N 15
PCIe_RX_P 16
MIPI0_D0_P 17
PCIe_RX_N 18
GND 19
GND 20
MIPI0_D1_N 21
PCIe_TX_P 22
MIPI0_D1_P 23
PCIe_TX_N 24
GND 25
GND 26
MIPI0_CLK_N 27
USB3-0-RX_N 28
MIPI0_CLK_P 29
USB3-0-RX_P 30
GND 31
GND 32
MIPI0_D2_N 33
USB3-0-D_P 34
MIPI0_D2_P 35
USB3-0-D_N 36
GND 37
GND 38
MIPI0_D3_N 39
USB3-0-TX_N 40
MIPI0_D3_P 41
USB3-0-TX_P 42
GND 43
44
45
46
HDMI1_SDA 47
HDMI1_TX2_P 48
HDMI1_SCL 49
HDMI1_TX2_N 50
HDMI1_CEC 51
GND 52
HDMI0_CEC 53
HDMI1_TX1_P 54
HDMI0_HOTPLUG 55
HDMI1_TX1_N 56
GND 57
GND 58
USB3-1-RX_N 59
HDMI1_TX0_P 60
USB3-1-RX_P 61
HDMI1_TX0_N 62
GND 63
GND 64
USB3-1-DP 65
HDMI1_CLK_P 66
USB3-1-DM 67
HDMI1_CLK_N 68
GND 69
GND 70
USB3-1-TX_N 71
HDMI0_TX2_P 72
USB3-1-TX_P 73
HDMI0_TX2_N 74
GND 75
GND 76
MIPI1_D0_N 77
HDMI0_TX1_P 78
MIPI1_D0_P 79
HDMI0_TX1_N 80
GND 81
GND 82
MIPI1_D1_N 83
HDMI0_TX0_P 84
MIPI1_D1_P 85
HDMI0_TX0_N 86
GND 87
GND 88
MIPI1_D1_P 89
MIPI1_C_N 90
HDMI0_CLK_P 91
MIPI1_C_P 92
HDMI0_CLK_N 93
GND 94
GND 95
MIPI1_D2_N 96
MIPI1_D3_N 97
MIPI1_D2_P 98
MIPI1_D3_P 99
GND 100
HDMI0_SDA 101
HDMI0_SCL 102
GND
GND
GND
GND
GND
GND
10164227-1004A1RLF


TF1
SD_DAT2 1
SD_DAT3 2
SD_CMD 3
SD_3.3V 4
NetR12_2 5
6
DAT2 7
DAT3 8
CMD 9
VDD 10
CLK 11
GND 12
DAT0 SD_DAT0
DAT1 SD_DAT1
DETECT1
DETECT2
SHIELD
GND
5033981892
USB0 connector-local ESD (separate from USB3 RX)


U6
TPD4EUSB30DQAR
USB3-0-RX_N 10
USB3-0-RX_N 1
USB3-0-RX_P 2
USB3-0-RX_P 9
NC
I/O1
I/O2
NC
GND
NC
I/O3
I/O4
NC
GND
GND
3
8
GND
GND


U16
TPD4EUSB30DQAR
USB0_N
USB0_P
NC
I/O1
I/O2
NC
GND
NC
I/O3
I/O4
NC
GND
GND
3
8
GND
GND


5V
C8
10uF
GND


Type_C
USB4105-GF-A
VBUS A4 5V
CC1 A5 CC1
CC2 B5 CC2
D- A7 USB0_N
D- B7 USB0_N
D+ A6 USB0_P
D+ B6 USB0_P
SBU1 A8
SBU2 B8
SHIELD
GND
SH
A1
GND
GND

MIPI0 protection

MIPI1 protection

GPIO_VREF GPIO_VREF

0R R16 CM4_3V3

R17 CM4_1V8

U5

TPD4EUSB30DQAR

MIPI0_D0_N 10 NC — NC 7 MIPI0_D1_N
MIPI0_D0_N 1 I/O1 — I/O3 4 MIPI0_D1_N
MIPI0_D0_P 2 I/O2 — I/O4 5 MIPI0_D1_P
MIPI0_D0_P 9 NC GND GND NC 6 MIPI0_D1_P
3 GND
8 GND

U8

TPD4EUSB30DQAR

MIPI1_D0_N 10 NC — NC 7 MIPI1_D1_N
MIPI1_D0_N 1 I/O1 — I/O3 4 MIPI1_D1_N
MIPI1_D0_P 2 I/O2 — I/O4 5 MIPI1_D1_P
MIPI1_D0_P 9 NC GND GND NC 6 MIPI1_D1_P
3 GND
8 GND

5V C26 10uF GND

5V C27 100nF GND

U7

TPD4EUSB30DQAR

MIPI0_CLK_N 10 NC — NC 7 MIPI0_D2_N
MIPI0_CLK_N 1 I/O1 — I/O3 4 MIPI0_D2_N
MIPI0_CLK_P 2 I/O2 — I/O4 5 MIPI0_D2_P
MIPI0_CLK_P 9 NC GND GND NC 6 MIPI0_D2_P
3 GND
8 GND

U9

TPD4EUSB30DQAR

MIPI1_CLK_N 10 NC — NC 7 MIPI1_D2_N
MIPI1_CLK_N 1 I/O1 — I/O3 4 MIPI1_D2_N
MIPI1_CLK_P 2 I/O2 — I/O4 5 MIPI1_D2_P
MIPI1_CLK_P 9 NC GND GND NC 6 MIPI1_D2_P
3 GND
8 GND

U14

TPD4EUSB30DQAR

MIPI0_D3_N 10 NC — NC 7 CAM_GPIO0
MIPI0_D3_N 1 I/O1 — I/O3 4 CAM_GPIO0
MIPI0_D3_P 2 I/O2 — I/O4 5 CAM_GPIO1
MIPI0_D3_P 9 NC GND GND NC 6 CAM_GPIO1
3 GND
8 GND

U10

TPD4EUSB30DQAR

MIPI1_D3_N 10 NC — NC 7 ID_SC
MIPI1_D3_N 1 I/O1 — I/O3 4 ID_SC
MIPI1_D3_P 2 I/O2 — I/O4 5 ID_SD
MIPI1_D3_P 9 NC GND GND NC 6 ID_SD
3 GND
8 GND

C18 100nF MIPI_3.3V

GPIO_VREF 1.8k CM-R2 ID_SC

GPIO_VREF 1.8k CM-R3 ID_SD

C20 100nF MIPI_3.3V

J1

MIPI0_D0_N 22 GND — GND 10 MIPI0_D3_N
MIPI0_D0_P 21 D0_N — D3_N 9 MIPI0_D3_P
GND 20 D0_P — D3_P 8 GND
MIPI0_D1_N 19 GND — GND 7
MIPI0_D1_P 18 D1_N — GPIO0 6 CAM_GPIO0
GND 17 D1_P — GPIO1 5 CAM_GPIO1
MIPI0_CLK_N 16 GND — GND 4
MIPI0_CLK_P 15 CLK_N — SCL 3 SCL0
GND 14 CLK_P — SDA 2 SDA0
MIPI0_D2_N 13 GND — 3V3 1 MIPI_3.3V
MIPI0_D2_P 12 D2_N — SHIELD MP
GND 11 D2_P — GND
FH12-22S-0.5SVA(54)

J3

MIPI1_D0_N 22 GND — GND 10 MIPI1_D3_N
MIPI1_D0_P 21 D0_N — D3_N 9 MIPI1_D3_P
GND 20 D0_P — D3_P 8 GND
MIPI1_D1_N 19 GND — GND 7
MIPI1_D1_P 18 D1_N — GPIO0 6 CAM_GPIO0
GND 17 D1_P — GPIO1 5 CAM_GPIO1
MIPI1_CLK_N 16 GND — GND 4
MIPI1_CLK_P 15 CLK_N — SCL 3 ID_SC
GND 14 CLK_P — SDA 2 ID_SD
MIPI1_D2_N 13 GND — 3V3 1 MIPI_3.3V
MIPI1_D2_P 12 D2_N — SHIELD MP
GND 11 D2_P — GND
FH12-22S-0.5SVA(54)

PI1

MIPI_3.3V 1 MIPI_3.3V — ID_SC 28 ID_SC
GPIO15 10 GPIO15 — GPIO5 29 GPIO5
GPIO17 11 GPIO17 — GPIO2 3 GPIO2
GPIO18 12 GPIO18 — GND 30
GPIO27 13 GPIO27 — GPIO6 31 GPIO6
14 GND — GPIO12 32 GPIO12
GPIO22 15 GPIO22 — GPIO13 33 GPIO13
GPIO23 16 GPIO23 — GND 34
MIPI_3.3V 17 MIPI_3.3V — GPIO19 35 GPIO19
GPIO24 18 GPIO24 — GPIO16 36 GPIO16
GPIO10 19 GPIO10 — GPIO26 37 GPIO26
5V 2 5V — GPIO20 38 GPIO20
20 GND — GND 39
GPIO9 21 GPIO9 — 5V 4 5V
GPIO25 22 GPIO25 — GPIO21 40 GPIO21
GPIO11 23 GPIO11 — GPIO3 5 GPIO3
GPIO8 24 GPIO8 — GND 6
25 GND — GPIO4 7 GPIO4
GPIO7 26 GPIO7 — GPIO14 8 GPIO14
ID_SD 27 ID_SD — GND 9
GND
TSM-120-01-T-DV

Power declarations: USB-C supplies 5V and GND;
GPIO_VREF is supplied through the fitted H2 shunt (default 2-3).

PWR_FLAG 5V

PWR_FLAG GND

PWR_FLAG GPIO_VREF

Sheet: /
File: cm5-nano-compatible.kicad_sch
Title: